GENERATIVE SYNTHETIC DATA GENERATION IN HEALTHCARE

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***Abstract*—The scarcity of accessible, high-quality healthcare data due to privacy constraints has driven the need for synthetic alternatives. This work investigates the use of generative models—specifically GANs and VAEs—for producing realistic synthetic healthcare data. We assess the generated data in terms of statistical similarity, utility for model training, and privacy preservation. Results demonstrate that generative synthetic data can serve as a viable substitute for real patient data in machine learning pipelines, supporting research and innovation while adhering to data protection norms.**

I. Introduction

In recent years, the healthcare sector has rapidly embraced data-centric technologies to enhance patient outcomes, streamline diagnostics, and improve operational workflows. Despite this growth, the utilization of real-world patient information is often limited due to strict privacy mandates such as the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR). These constraints create significant hurdles in acquiring the vast and diverse datasets required for training high-performance machine learning and deep learning models.

To overcome these limitations, there has been a growing focus on the use of generative synthetic data—artificially generated data that preserves the underlying statistical features of original datasets while safeguarding personal information. Leveraging deep learning frameworks such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), these techniques are capable of learning intricate patterns in healthcare data and producing realistic yet anonymized replicas suitable for research and development.

From an information technology perspective, the creation of synthetic healthcare data offers a balanced approach to maintaining data confidentiality while enabling innovation. It allows for the ethical sharing of data, encourages academic and institutional collaboration, and supports the development of more accurate and generalizable AI models. This paper presents an in-depth discussion on the generation, application, and implications of synthetic data in healthcare, underscoring its role as a viable, privacy-preserving alternative to real patient records.

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II. Related Work

The application of Generative Adversarial Networks (GANs) for synthetic data generation has gained significant traction across various healthcare and industrial domains. In one study, standard GANs were employed to augment liver lesion datasets, achieving notable improvements with a sensitivity of 85.7 and specificity of 92 [1].4. Another approach introduced the Bias-transforming GAN (Bt-GAN), which focused on generating fairer synthetic healthcare data. Although the study did not disclose exact performance metrics, it reported enhanced accuracy and fairness [2]. Similarly, medGAN, a hybrid model combining autoencoders and GANs, was developed to produce discrete multi-label patient records. This method showed strong resemblance to real patient data while minimizing privacy risks [3]. For medical image generation, MI-GAN was used to synthesize retinal images, reporting Dice coefficients of 0.837 on the STARE dataset and 0.832 on DRIVE, indicating effective segmentation performance [4]. Further advancements included a divide-and-conquer-enhanced CTGAN model for generating synthetic tabular data, which achieved an AUC of 85.2 and an F1-score of 74.78 for non-small cell lung cancer (NSCLC), with performance varying across other datasets [5]. In another notable work, GAN-generated synthetic data was paired with the EfficientNet-B7 model to aid breast cancer diagnosis, resulting in an accuracy of 92.01, F1-score of 92.01, and a precision of 92.07 [6]. CycleGAN was applied for data augmentation in medicine and neuroscience, achieving excellent results with 97.3 accuracy, an F1-score of 0.946, and an AUC of 0.992 [7]. In the domain of synthetic media generation, an improved GAN architecture incorporating label smoothing and data augmentation achieved an FID score of 55.67, with a remarkable accuracy of 98.82 and F1-score of 0.99 [8]. Addressing limitations in medical imaging, the Multi-Atlas Image Synthesis (MAISI) method demonstrated effectiveness with an average FID score of 2.308 [9]. Finally, GANs were integrated with Random Forest classifiers for burst failure risk analysis in industrial pipelines, enhancing prediction capabilities when supplemented with synthetic data, though exact performance values were not detailed [10].

III. Objectives of This Study

This paper aims to provide a comprehensive examination of the emerging role of generative synthetic data in the healthcare domain. The key goals of this research are outlined below:

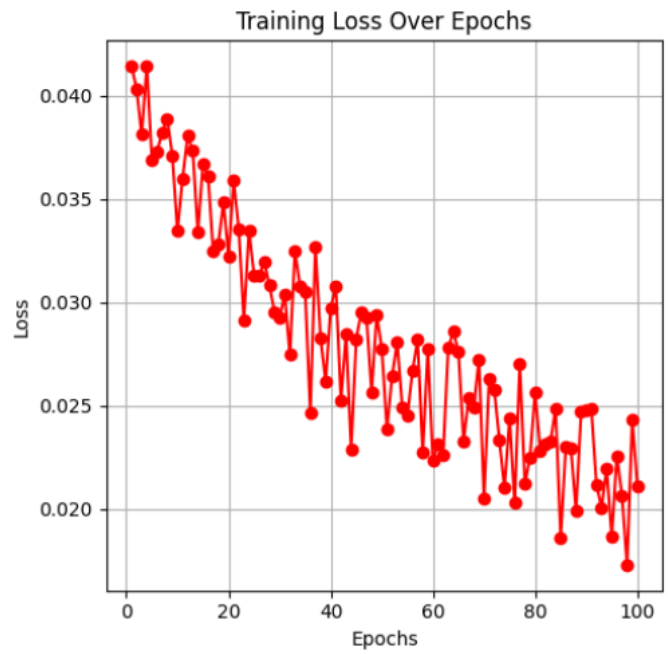
Examine the Relevance of Synthetic Data in Healthcare To understand the foundational concept of synthetic data and emphasize its significance in modern healthcare systems. Special attention is given to how synthetic data enables research and technological development while ensuring patient privacy remains protected.

Analyze Generative Techniques for Data Creation To study a range of generative models—such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and other advanced deep learning methods—used in producing synthetic healthcare data that closely mirrors the statistical characteristics of real-world datasets. Assess the Practical Value of Synthetic Data To evaluate how accurately synthetic data replicates meaningful trends and insights present in original datasets, and to determine its effectiveness for applications like disease prediction, medical diagnostics, and personalized treatment strategies.

Explore Challenges and Barriers To identify and discuss the technical complexities, ethical concerns, and legal constraints currently affecting the adoption of synthetic data in healthcare, especially in live clinical environments.

Suggest Future Research Directions To propose areas for further innovation, including ways to improve the quality, reliability, and ethical usage of synthetic data, thereby promoting its wider adoption and enhancing its impact on healthcare transformation.

A. *The Role of Generative Models in Modern Healthcare*



In the era of digital healthcare, data has become a fundamental asset in driving innovation, enhancing patient care, and developing intelligent decision-support systems. With the rapid integration of artificial intelligence and machine learning into clinical workflows, the demand for large, diverse, and high-quality medical datasets has surged. However, real patient data is often tightly regulated due to legal, ethical, and privacy-related concerns, making access and sharing extremely challenging. This has created a pressing need for alternative approaches that can maintain data utility while safeguarding patient identities. Synthetic data generation, particularly through generative modeling techniques, offers a viable and promising solution to this dilemma. Unlike traditional anonymization, synthetic data is created from scratch using mathematical models trained on real datasets, ensuring that no actual patient information is exposed. This paradigm not only addresses privacy issues but also fosters greater collaboration between institutions, enables safe data sharing, and supports scalable experimentation. By leveraging advanced models such as GANs and VAEs, researchers can generate synthetic datasets that closely resemble real medical data in structure and statistical properties, making them suitable for training, validating, and testing AI-based healthcare applications. This paper delves into the growing importance of generative synthetic data, investigates current methodologies, and explores how these technologies are shaping the future of secure and ethical data-driven healthcare systems.

B. Model Performance

The Fully Connected Neural Network (FCNN) employed in this study demonstrated strong performance across multiple evaluation metrics. The model achieved an accuracy of 98.34 Percentage, indicating its high capability in correctly classifying data instances. Furthermore, it recorded a precision of 96.59, reflecting its effectiveness in minimizing false positives. The recall stood at 96.34, suggesting a low rate of false negatives, and the overall F1-score was 96.30, showcasing a strong balance between precision and recall.

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